

UNIVERZA V LJUBLJANI
FAKULTETA ZA RAČUNALNIŠTVO IN INFORMATIKO

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**Ocenjevanje volatilnosti z
Markovskim prehajanjem na
finančnih trgih s sekvenčnimi Monte
Carlo metodami**

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**Estimating Markov switching
volatility in financial markets using
sequential Monte Carlo methods**

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THE 2ND CYCLE MASTER'S STUDY PROGRAMME
COMPUTER AND INFORMATION SCIENCE

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Vito Levstik, 2026

To all the flowers of this world.

*"The only reason for time is so that
everything doesn't happen at once."*

— Albert Einstein

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List of used acronyms

acronym	meaning
CA	classification accuracy
DBMS	database management system
SVM	support vector machine
...	...

Abstract

Title: Estimating Markov switching volatility in financial markets using sequential Monte Carlo methods

This sample document presents an approach to typesetting your BSc thesis using L^AT_EX. A proper abstract should contain around 100 words which makes this one way too short. A good abstract contains: (1) a short description of the tackled problem, (2) a short description of your approach to solving the problem, and (3) (the most successful) result or contribution in your thesis.

Keywords

computer, computer, computer

Povzetek

Naslov: Ocenjevanje volatilnosti z Markovskim prehajanjem na finančnih trgih s sekvenčnimi Monte Carlo metodami

V vzorcu je predstavljen postopek priprave magistrskega dela z uporabo okolja L^AT_EX. Vaš povzetek mora sicer vsebovati približno 100 besed, ta tukaj je odločno prekratek. Dober povzetek vključuje: (1) kratek opis obravnavanega problema, (2) kratek opis vašega pristopa za reševanje tega problema in (3) (najbolj uspešen) rezultat ali prispevek magistrske naloge.

Ključne besede

računalnik, računalnik, računalnik

Razširjeni povzetek

To je primer razširjenega povzetka v slovenščini, ki je obvezen za naloge pisane v angleščini. Razširjeni povzetek mora vsebovati vse glavne elemente dela napisanega v angleščini skupaj s kratkim uvodom in povzetkom glavnih elementov metode, glavnih eksperimentalnih rezultatov in glavnih ugotovitev. Razširjeni povzetek naj bo strukturiran v podpoglavja (spodaj je naveden le okvirni primer in je nezavezujoč). Čez palec navadno razširjeni povzetek nanese okoli 10 odstotkov obsega celotnega dela.

I Kratek pregled sorodnih del

II Predlagana metoda

III Eksperimentalna evaluacija

IV Sklep

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Chapter 1

Introduction

Chapter 2

Stochastic Volatility

Volatility is a fundamental concept in financial econometrics and is commonly used as a measure of the variability or uncertainty of asset returns. In its most basic form, volatility refers to the standard deviation of returns over a given time horizon, and therefore quantifies the magnitude of fluctuations around the expected return.

Many models assume that volatility is constant. In practice, however, financial markets exhibit periods of high volatility, such as during financial crises, and periods of low volatility, such as during stable economic conditions. Stochastic volatility models capture this behavior by allowing volatility to vary over time according to a stochastic process.

This chapter begins by introducing volatility and then develops the state space formulation used throughout the thesis, with particular emphasis on the Markov switching stochastic volatility (MSSV) model.

2.1 Volatility

Let P_t denote the price of an asset (stock, index, commodity, etc.) at time t . The log return of the asset is defined as

$$Y_t = \log(P_t) - \log(P_{t-1}) = \log\left(\frac{P_t}{P_{t-1}}\right). \quad (2.1)$$

The advantage of using log returns rather than simple returns lies in their additive structure over time and their natural treatment of compounding effects [1]. Throughout this thesis, the term return refers to a log return.

While returns are directly observable, volatility is a characteristic of the underlying return distribution and is not directly observable from a single realization. At each time point t , only one return Y_t is observed, whereas volatility is defined through the variability of the entire return-generating distribution. Consequently, time-varying volatility must be inferred from data rather than measured directly.

To formalize this idea, let $\mathcal{F}_{t-1}$ denote the information available up to time $t - 1$. The conditional mean and variance of the return are defined as

$$\mu_t = \mathbb{E}[Y_t | \mathcal{F}_{t-1}], \quad (2.2)$$

$$\sigma_t^2 = \text{Var}(Y_t | \mathcal{F}_{t-1}) = \mathbb{E}[(Y_t - \mu_t)^2 | \mathcal{F}_{t-1}]. \quad (2.3)$$

The quantity σ_t is referred to as the conditional volatility. In many financial applications, the conditional mean is assumed to be negligible relative to the variability of returns, so that the focus is primarily on modelling the conditional volatility process.

When high-frequency intraday data are available, volatility can be estimated nonparametrically using realized volatility. Suppose that within a fixed time interval, for example one trading day, we observe intraday log returns $r_{t,i}$ for $i = 1, \dots, m$. The realized variance is defined as

$$RV_t = \sum_{i=1}^m r_{t,i}^2, \quad (2.4)$$

Under certain regularity conditions, as the sampling frequency increases (i.e., $m \rightarrow \infty$), the realized variance converges to the quadratic variation of the log-price process, which in the absence of jumps corresponds to the integrated variance over the interval. This provides a theoretical justification for using realized volatility as a consistent estimator of latent daily volatility.

2.2 State space models

A state space model describes a dynamical system in which an unobserved (latent) stochastic process drives the evolution of an observed process. The central idea is that the system evolves through hidden states, while only noisy measurements of those states are available.

Let $\{X_t\}_{t \geq 0}$ denote the sequence of random variables representing the latent state process, $\{Y_t\}_{t \geq 1}$ denote the sequence of random variables representing the observed process and let T denote the time horizon, hence $t \in \{0, 1, \dots, T\}$ ¹. X_t takes values in a measurable state space $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$ and Y_t takes values in $(\mathcal{Y}, \mathcal{B}(\mathcal{Y}))$. Together, $\{(X_t, Y_t)\}_{t \geq 1}$ forms a stochastic process on the joint space $(\mathcal{X} \times \mathcal{Y}, \mathcal{B}(\mathcal{X}) \otimes \mathcal{B}(\mathcal{Y}))$ [2].

The latent process $\{X_t\}_{t \geq 0}$ is modelled as a Markov process, while the observations $\{Y_t\}_{t \geq 1}$ are conditionally independent given the state process. The model is specified by the following densities:

$$X_0 \sim \chi(x_0|\theta), \quad (2.5)$$

$$X_t|X_{t-1} = x_{t-1} \sim p(x_t|x_{t-1}, \theta), \quad (2.6)$$

$$Y_t|X_t = x_t \sim q(y_t|x_t, \theta), \quad (2.7)$$

where χ is the *initial density*, p is the *transition density*, and q is the *observation* or *emission density*. The vector $\theta \in \Theta$ denotes the collection of model parameters [2, 3, 4].

A potential problem with this definition is that conditional distribution of X_t given X_{t-1} may not always admit a density. A proper way would be to define the state space model in a more measure theoretic way using probability kernels [2]. However, we will not deal with this more general definition in this thesis, as our models will admit nice densities.

¹We indexed the latent states from 0 to T , whereas the observed returns are indexed from 1 to T . This way we want to emphasize that the initial state X_0 is the prior and we don't associate it with any observation. One could define the indexing differently (e.g. observations also starting with $t = 0$), but we decided to use this convention.

Sometimes, the state space model is defined in even more general terms, where the latent process and observation process are also dependent on the control input U_t , which constitutes another sequence of random variables [5]. Such definition is more used in control theory and engineering. The control inputs can be thought of as external factors that influence the system dynamics and observations. In this thesis, we will not consider control inputs, as we are primarily interested in modelling the latent volatility process and its relationship with observed returns.

The main goal of state space models is to estimate the latent states based on the observed data. With $X_{i:j} = \{X_i, X_{i+1}, \dots, X_j\}$ we denote the sequence of random variables from time i to j , and with $x_{i:j} = \{x_i, x_{i+1}, \dots, x_j\}$ we denote its realization. From joint posterior distribution of the latent states and model parameters given the observed data $p(x_{0:t}, \theta \mid y_{1:t})$, we can compute the following marginal distributions of interest:

- The *smoothing distribution*

$$p(x_{0:t} \mid y_{1:t}) = \int p(x_{0:t}, \theta \mid y_{1:t}) d\theta \quad (2.8)$$

- The *filtering distribution*

$$p(x_t \mid y_{1:t}) = \int p(x_{0:t}, \theta \mid y_{1:t}) dx_{0:t-1} d\theta \quad (2.9)$$

- The *prediction distribution*

$$p(x_{t+1} \mid y_{1:t}) = \int p(x_{t+1} \mid x_t, \theta) p(x_{0:t}, \theta \mid y_{1:t}) dx_{0:t} d\theta \quad (2.10)$$

- The *posterior distribution of the parameters*

$$p(\theta \mid y_{1:t}) = \int p(x_{0:t}, \theta \mid y_{1:t}) dx_{0:t} \quad (2.11)$$

A special case of state space models are the hidden Markov models (HMM). Although there is some discrepancy in the literature, the term hidden Markov model is often used to refer to a state space model in which the latent state is discrete and takes values in a finite set $\{1, 2, \dots, K\}$, where K is the number of states [2].

2.2.1 Linear Gaussian state space model

An important class of state space models is the linear Gaussian state space model (LG-SSM), in which both the transition and observation equations are linear and Gaussian. Mathematically, the model is defined as

$$X_t = A_t X_{t-1} + B_t U_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, Q_t), \quad (2.12)$$

$$Y_t = C_t X_t + D_t U_t + \delta_t, \quad \delta_t \sim \mathcal{N}(0, R_t), \quad (2.13)$$

where A_t, B_t, C_t, D_t are matrices that define the dynamics, and Q_t and R_t are covariance matrices of the process and observation noise, respectively. Note that if matrices A_t, B_t, C_t, D_t are time independent, the model is said to be stationary [5]. Here, we included the control input U_t in the model, as it is commonly used in the literature. However, as mentioned in 2.2, we will not consider control inputs in this thesis, so we omit them.

Using our constraints we can write the model in a more density focused notation

$$X_t | X_{t-1} = x_{t-1} \sim \mathcal{N}(A_t x_{t-1}, Q_t), \quad (2.14)$$

$$Y_t | X_t = x_t \sim \mathcal{N}(C_t x_t, R_t). \quad (2.15)$$

Let θ denote the collection of model parameters. If at time $t = 0$, the initial state is Gaussian as well

$$X_0 \sim \mathcal{N}(\mu_0, Q_0), \quad (2.16)$$

then all subsequent filtering distributions $p(x_t | y_{1:t}, \theta)$ are also Gaussian. Using the algorithm known as the Kalman filter, we can compute the parameters of these Gaussian filtering distributions in a closed form [5].

2.3 Markov switching stochastic volatility model

The Markov switching stochastic volatility (MSSV) model is the main model of interest in this thesis. It is a state space model in which the hidden state

is the tuple $X_t = (H_t, S_t)$. The log volatility H_t is continuous, while the regime indicator S_t is discrete and takes values in $\{1, 2, \dots, K\}$, where K is the number of regimes [6]. The latent state space is therefore

$$\mathcal{X} = \mathbb{R} \times \{1, 2, \dots, K\}, \quad (2.17)$$

equipped with the product σ -algebra $\mathcal{B}(\mathbb{R}) \otimes 2^{\{1, 2, \dots, K\}}$.

- **Observation equation:**

The observation distribution is defined with

$$Y_t | X_t = (h_t, s_t) \sim \mathcal{N}(0, \exp(h_t)), \quad (2.18)$$

where $\mathcal{N}$ denotes the normal distribution with mean and variance [6, 7]. Such distribution admits a density function which we denote by $q(y_t | x_t, \theta) = q(y_t | h_t)$, as the observation density does not depend on the regime s_t or any other model parameters θ .

Note that mathematically speaking the latent state H_t is the log variance of the return Y_t , but it is often referred to as the log volatility in the literature [7, 8, 9]. We will also follow this convention and refer to H_t as the log volatility.

- **Transition equation:**

The transition distribution is defined with

$$H_t | H_{t-1} = h_{t-1}, S_t = s_t \sim \mathcal{N}(\mu_{s_t} + \phi(h_{t-1} - \mu_{s_t}), \eta^2), \quad (2.19)$$

$$S_t | S_{t-1} = s_{t-1} \sim \text{Categorical}([p_{s_{t-1}, 1}, \dots, p_{s_{t-1}, K}]), \quad (2.20)$$

where μ_{s_t} denotes the regime-specific mean associated with state s_t , ϕ is the persistence parameter, η is the standard deviation of the log-volatility innovation process [6]. The values $p_{ij} = \mathbb{P}(S_t = j | S_{t-1} = i)$ are the probabilities of transitioning from regime i to regime j , and

they are collected in the $K \times K$ regime transition matrix

$$P = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1K} \\ p_{21} & p_{22} & \cdots & p_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ p_{K1} & p_{K2} & \cdots & p_{KK} \end{bmatrix}.$$

Note that the rows of P must sum to 1, i.e., $\sum_{j=1}^K p_{ij} = 1$ for all i . Observe that the log volatility process is essentially an AR(1) process with regime-specific means.

At first it seems that such conditional definition breaks the Markov property of the latent state process, as H_t depends on S_t . However, the latent state process is still Markovian, as the transition distribution of the joint state (H_t, S_t) only depends on the previous joint state (H_{t-1}, S_{t-1}) . This can be seen by factoring the transition distribution as

$$p(x_t | x_{t-1}, \theta) = p((h_t, s_t) | (h_{t-1}, s_{t-1}), \theta) \quad (2.21)$$

$$= p(s_t | h_{t-1}, s_{t-1}, \theta) p(h_t | h_{t-1}, s_{t-1}, s_t, \theta) \quad (2.22)$$

$$= p(s_t | s_{t-1}, \theta) p(h_t | h_{t-1}, s_t, \theta), \quad (2.23)$$

- **Initial distribution:**

The MSSV model is completed by specifying the initial distribution as

$$H_0 | S_0 = s_0 \sim \mathcal{N}\left(\mu_{s_0}, \frac{\eta^2}{(1-\phi^2)}\right), \quad (2.24)$$

$$S_0 \sim \text{Categorical}\left(\frac{1}{K}, \frac{1}{K}, \dots, \frac{1}{K}\right), \quad (2.25)$$

according to the stationary distribution of the corresponding AR(1) component, with mean μ_{s_0} and variance $\frac{\eta^2}{(1-\phi^2)}$, and a uniform prior over the regimes, respectively [6].

The initial distribution admits a density function which we can write out as $\chi(x_0 | \theta) = \chi(s_0 | \theta) \chi(h_0 | s_0, \theta)$.

In order for the MSSV model to be well-defined, the following assumptions are made:

1. The persistence ϕ lies in the range $(-1, 1)$.
2. Each regime should correspond to at least one time point.
3. The regime means are ordered such that $\mu_1 < \mu_2 < \dots < \mu_K$.

Assumption 1 ensures that the AR(1) process is stationary. This is needed to ensure that the log volatility does not diverge to infinity. Assumption 2 is necessary to avoid empty regimes, otherwise some μ_i would be unidentified. Assumption 3 is a common identification constraint in mixture models, which prevents label switching and ensures that the regimes are uniquely defined [6].

In this thesis we ensure the regime identifiability using following parameterization:

$$\mu_i = \mu_{i-1} + e^{\delta_i}, \quad i = 2, \dots, K, \quad (2.26)$$

where $\delta_i \in \mathbb{R}$. Such parameterization is motivated by the fact that the exponential function is strictly positive, which guarantees that $\mu_i > \mu_{i-1}$ and leaves δ_i unconstrained.

The parameters of the MSSV model are then given by vector

$$\theta = \{\mu_1, \delta_2, \dots, \delta_K, \phi, \eta, P\}.$$

Since P is row stochastic, we have $K(K-1)$ free parameters in P . Therefore, the total number of free parameters is $K + 2 + K(K-1) = K^2 + 2$.

Note that in case of $K = 1$, the MSSV model reduces to the standard stochastic volatility (SV) model, in which only the log volatility H_t is a latent variable and there are no regimes.

2.4 Problem formulation

The goal of this thesis is to estimate volatility in financial markets, using the MSSV model with parameters θ . Let $y_{1:t} = \{y_1, y_2, \dots, y_t\}$ de-

note the sequence of observed returns up to time $t \leq T$. Similarly, let $x_{0:t} = \{x_0, x_1, \dots, x_t\}$ denote the sequence of latent states up to time t , where x_0 is the initial prior state. In general we can distinguish between two types of our problem

1. Known parameters θ

If we assume that the model parameters θ are known, then the main objective is to estimate

$$p(x_{0:t} \mid y_{1:t}, \theta), \quad (2.27)$$

which we can expand using Bayes' theorem:

$$p(x_{0:t} \mid y_{1:t}, \theta) \propto p(y_{1:t} \mid x_{0:t}, \theta) p(x_{0:t} \mid \theta). \quad (2.28)$$

We can further expand the likelihood:

$$p(y_{1:t} \mid x_{0:t}, \theta) \propto q(y_t \mid x_{0:t}, y_{1:t-1}, \theta) p(y_{1:t-1} \mid x_{0:t}, \theta) \quad (2.29)$$

$$\propto q(y_t \mid x_t, \theta) q(y_{t-1} \mid x_{0:t}, y_{1:t-2}, \theta) p(y_{1:t-2} \mid x_{0:t}, \theta) \quad (2.30)$$

$$\propto \prod_{i=1}^t q(y_i \mid x_i, \theta), \quad (2.31)$$

$$= \prod_{i=1}^t q(y_i \mid x_i), \quad (2.32)$$

where we used the fact that observations y_i depend only on the current latent state x_i (actually only on h_i) and not on the entire history of latent states, observations, or parameters. We can also expand:

$$p(x_{0:t} \mid \theta) \propto p(x_t \mid x_{0:t-1}, \theta) p(x_{0:t-1} \mid \theta) \quad (2.33)$$

$$\propto p(x_t \mid x_{t-1}, \theta) \dots p(x_1 \mid x_0, \theta) \chi(x_0 \mid \theta) \quad (2.34)$$

$$= \prod_{i=1}^t p(x_i \mid x_{i-1}, \theta) \chi(x_0 \mid \theta), \quad (2.35)$$

where we used the Markov property of the latent state process.

The final expression is then given by

$$p(x_{0:t} \mid y_{1:t}, \theta) \propto \chi(x_0 \mid \theta) \prod_{i=1}^t p(x_i \mid x_{i-1}, \theta) q(y_i \mid x_i). \quad (2.36)$$

2. Unknown parameters θ

In reality we do not know the model parameters θ of our MSSV model, so we would also like to estimate them. The main objective is then the joint posterior distribution of the latent states and model parameters given the observed data:

$$p(x_{0:t}, \theta \mid y_{1:t}), \quad (2.37)$$

from which we can compute the smoothing, filtering and prediction distributions of latent states and the posterior distribution of the parameters as described in Section 2.2.

Let's expand the joint posterior distribution using Bayes' theorem:

$$p(x_{0:t}, \theta \mid y_{1:t}) = \frac{p(x_{0:t}, y_{1:t} \mid \theta) p(\theta)}{p(y_{1:t})} \quad (2.38)$$

$$\propto p(y_{1:t} \mid x_{0:t}, \theta) p(x_{0:t} \mid \theta) p(\theta). \quad (2.39)$$

Reusing the previously derived expressions, the joint posterior distribution can then be expressed as

$$p(x_{0:t}, \theta \mid y_{1:t}) \propto p(\theta) \chi(x_0 \mid \theta) \prod_{i=1}^t p(x_i \mid x_{i-1}, \theta) q(y_i \mid x_i). \quad (2.40)$$

From both Equations (2.36) and (2.40), we can see the sequential nature of the problem. If we somehow initially sample the prior latent state x_0 (and model parameters θ if they are unknown), we can sequentially propagate the latent states through transition density and update them with the new observations through the observation density. This is the main idea behind sequential Monte Carlo methods, which we will present in the next chapter.

2.4.1 Prior distributions of the parameters

In the case of unknown model parameters θ , we need to specify their prior distributions. We assume that the priors of the parameters are independent, so that the joint prior distribution can be factorized as

$$p(\theta) = p(\mu_1) \prod_{i=2}^K p(\delta_i) p(\phi) p(\eta) p(P). \quad (2.41)$$

Even though we defined our model with parameters δ_i and ϕ , we will define the priors in terms of $\Delta_i = \mu_i - \mu_{i-1} = e^{\delta_i}$ and $\varphi = \frac{\phi+1}{2}$. The reason for such reparameterization of ϕ to φ is that now φ lies in the range $(0, 1)$, which allows us to use a Beta distribution as a prior. The reason for reparameterization of δ_i to Δ_i is that now we can use a truncated normal distribution as a prior, which will be more convenient for conditional sampling later on.

We choose the following prior distributions for the parameters:

$$\mu_1 \sim \mathcal{N}(0, 5^2), \quad (2.42)$$

$$\Delta_i \sim \mathcal{N}_{(0, \infty)}(0, 5^2), \quad i = 2, \dots, K, \quad (2.43)$$

$$\varphi \sim \mathcal{B}(6, 2), \quad (2.44)$$

$$\eta \sim \mathcal{IG}(2.001, 0.1), \quad (2.45)$$

$$P_{i,:} \sim \text{Dirichlet}(\alpha_{i,1}, \dots, \alpha_{i,K}), \quad i = 1, \dots, K, \quad (2.46)$$

where $\mathcal{N}$ denotes the normal distribution, $\mathcal{N}_{(0, \infty)}$ denotes the truncated normal distribution on the positive half-line, $\mathcal{B}$ denotes the Beta distribution, $\mathcal{IG}$ denotes the inverse gamma distribution and Dirichlet denotes the Dirichlet distribution.

The hyperparameters of the prior distributions are chosen to be slightly informative. For μ_1 and Δ_i , the hyperparameters are such that cover quite a wide range of log volatilities, which translates to even bigger range after transformation to volatilities. For the prior of φ , the hyperparameters are such that the persistence parameter ϕ is likely to be positive, which is a common assumption in financial econometrics. For the prior of η , we follow

the combination of hyperparameters used in [7, 10]. The hyperparameters $\alpha_{i,j}$ of the Dirichlet prior are set to different values depending on the dataset. For synthetic datasets, we set $\alpha_{i,j} = 1.5$ for $i \neq j$ and $\alpha_{i,i} = 5$, which encourages the model to stay in the same regime, but still allows for regime switching. For real datasets, we set $\alpha_{i,j} = 1.5$ for $i \neq j$ and $\alpha_{i,i} = 21.5$.

Chapter 3

Sequential Monte Carlo Methods

Here we present the theory of sequential Monte Carlo (SMC) methods, starting with bootstrap particle filter (BPF) and then moving to more advanced methods such as the particle Monte Carlo Markov Chain (PMCMC) and SMC².

3.1 Bootstrap Particle Filter

3.2 PMCMC

3.3 SMC²

Chapter 4

Results

Chapter 5

Discussion

Chapter 6

Conclusion

Appendix A

Title of the appendix 1

Example of the appendix.

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